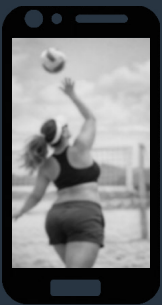
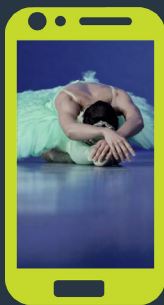
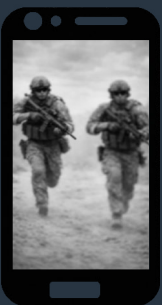
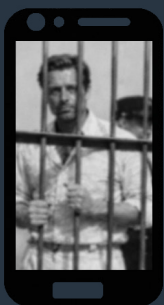
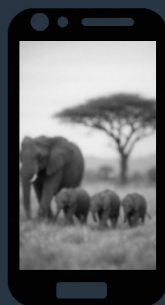
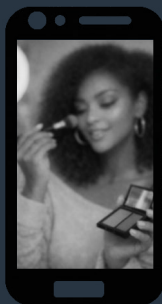
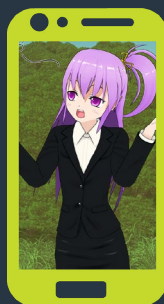
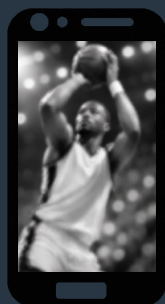


MOBILE PERSONALIZATION

The history and theory of
mobile streaming media



JUSTIN GRANDINETTI
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MOBILE PERSONALIZATION

*The History and Theory of
Mobile Streaming Media*

by
Justin Grandinetti &
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Mobile Personalization: The History and Theory of Mobile Streaming Media

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CHAPTER TWO

Architectures of Mobility

Introduction

You're standing in line at the airport, halfway through an episode of your favorite Netflix series on your phone. You didn't plan ahead, just opened the app and hit play. Around you, someone's watching TikTok clips on repeat, another is livestreaming a concert on Instagram, and a kid nearby is deep into a YouTube rabbit hole.

No wires, no downloads, no friction.

It feels effortless.

But behind that casual scroll or stream is an enormous and largely invisible machine. Your video didn't come from Netflix directly; it may have been served from a local cache server tucked into a telecom provider's infrastructure. Your phone switched between 5G and Wi-Fi, managed by complex traffic prioritization protocols. The app collected data about what you watched, how long you watched, and whether you paused or skipped. It wasn't just entertainment—it was a data transaction. This everyday scene illustrates our core argument: Mobile streaming's ease and personalization rest on complex, hidden infrastructures and platform systems working in concert. And this infrastructure didn't emerge overnight. It is instead the result of a long trajectory of sociotechnical changes.

This moment in the airport line isn't just a modern convenience. It's the front end of a vast technical and economic structure designed to deliver

media in motion—what we call *architectures of mobility*, the technical systems and economic platforms that make such personalized, mobile media flows possible. In this chapter, we move from historical antecedents of mobile streaming to the complex mix of infrastructural apparatuses that developed to make on-the-go interactions like the above situation work. This chapter unpacks what makes seamless mobile streaming possible, and what it costs.

Mobile streaming media has become a pervasive mode of cultural interaction enabled by the entanglement of platforms and infrastructures. We often think of streaming in terms of our personal devices, such as the ubiquitous smartphone or tablet, but the true architecture of mobile media extends far beyond handheld screens; it encompasses global wireless networks, cloud computing systems, data centers, content delivery networks, and platform ecosystems that deliver video, music, games, and other media to users while they are on the move. Manuel Castells, a key theorist of networked mobility, posits that communication networks create a space of flows wherein information travels instantly across vast distances, reconfiguring how we experience space and time.¹ The “architectures of mobility” are the technical systems and economic platforms that make such flows possible. This chapter examines the infrastructures and platforms enabling mobile streaming media in primarily Western contexts, analyzing how technical underpinnings (wireless networks, cloud infrastructure, big data pipelines, algorithms, and APIs) merge with platform business models to structure the contemporary media landscape. Rather than a single “streaming industry,” these services represent an infrastructural logic spanning multiple media and tech sectors.

As streaming has moved off the desktop and into pocket, the once-separate domains of infrastructure and platform have increasingly converged. Digital platforms like YouTube, Netflix, Spotify, or TikTok present themselves as content delivery services or social hubs for users, but they also operate as digital infrastructures in their own right.² Nick Srnicek defines platforms at the highest level as “digital infrastructures that enable two or more groups to interact.”³ In other words, a platform such as a streaming service not only hosts media content and connects users with that content, but also provides the foundational technical base—storage, bandwidth, interface, and data processing upon which those interactions run. At the same time,

¹ Manuel Castells, “Communication, Power and Counter-Power in the Network Society,” *International Journal of Communication* 1 (2007).

² Jean-Christophe Plantin et al., “Infrastructure Studies Meet Platform Studies in the Age of Google and Facebook,” *New Media & Society* 20, no. 1 (2018).

³ Nick Srnicek, *Platform Capitalism* (Polity Press, 2016).

traditional communications infrastructures have been reshaped by the logic of platforms. Jean-Christophe Plantin, Carl Lagoze, Paul Edwards, and Christian Sandvig suggest that “platform-based services acquire characteristics of infrastructure, while both new and existing infrastructures are built or reorganized on the logic of platforms.”⁴ In effect, companies like Google, Amazon, and Facebook increasingly build out physical infrastructure (such as fiber optic cables, server farms, and distributed content caches) to support their platforms, while infrastructure providers (like telecom carriers) adopt platform-like models of operation. The result is a layered system in which platform and infrastructure are deeply interwoven.

Crucially, mobile streaming architectures are not just technical systems; they are also economic and organizational systems. In effect, the platform has emerged as a dominant business model for the digital economy, one oriented around the extraction and use of data.⁵ Streaming media platforms exemplify this model; they function as intermediaries between content producers and audiences, positioning themselves in the middle of every interaction so as to capture data and monetize user’s activity.⁶ Whether supported by advertising (YouTube, TikTok), subscription fees (Netflix, Spotify), or hybrid models (Twitch’s subscriptions and ads), these services all leverage algorithmic tools to maximize engagement and profit. The very freedom and mobile personalization that mobile media promises to users, such as the ability to watch or listen to anything, anywhere, anytime, is underpinned by an elaborate apparatus of pervasive data collection operating behind the scenes. Of course, such participation in networked digital media often comes with a tradeoff: Users gain convenient access to information and communication at the expense of the ownership of their data, which is claimed by the operator of the enclosure.⁷ In other words, platforms enable the widespread collection of user data that becomes property of the owner of the platform in use. This captured information becomes a key resource for platforms, fueling the algorithms that recommend content and the targeted advertising or personalized experiences that drive revenue.

To uncover the material and digital infrastructure that enable mobile streaming, this chapter will unfold over four sections. First, we examine the platform business models that characterize streaming services, drawing on

⁴ Plantin et al., “Infrastructure Studies Meet Platform Studies,” 294.

⁵ Srnicek, *Platform Capitalism*.

⁶ Srnicek, *Platform Capitalism*.

⁷ Mark Andrejevic, “Surveillance and Alienation in the Online Economy,” *Surveillance & Society* 8, no. 3 (2010).

theorists of platforms as infrastructures to understand how mobile media platforms create value by aggregating content and users. We then delve into the wireless infrastructures that enable mobility—the evolution of cellular networks (3G, 4G, 5G) and Wi-Fi—and how these networks interface with streaming platforms to deliver high-bandwidth content on the go. Following that, we explore the technical underpinnings of contemporary streaming, such as the big data pipelines, algorithmic recommendation systems, and application programming interfaces (APIs) that allow platforms to mediate and customize the flow of media. Finally, we consider the convergence of platforms and infrastructures, how companies merge roles to control both content and its distribution, and discuss what these architectures of mobility mean for mobile media practices and users.

Platforms, Business Models, and the Data Economy of Streaming

Underneath the user-friendly interfaces of Netflix, YouTube, Spotify, TikTok, and other streaming services lies a distinctive business logic best described as “complex governance arrangements.” These arrangements include not only the platform’s API, which enables the API to parse through incredible amounts of data, but also the material conditions that imbue the platform with such power.⁸ Relatedly, the platform has emerged as a logic that became prominent in the 2000s, characterized by its ability to extract, control, and organize immense amounts of data. Platform-based companies seek to gather diverse users, such as individual consumers, advertisers, content producers, developers, etc., onto their digital infrastructures and facilitate interactions among them, thereby capturing value from each transaction or engagement. Data is the fuel that drives platform business logics, and it has proven to be wildly successful, as it acts almost as a renewable resource that a platform’s users top off with every click.

In the realm of mobile streaming media, this platform model manifests in several ways. Streaming services operate multi-sided markets: For example, YouTube connects content creators, viewers, and advertisers, whereas Spotify connects listeners, music rights holders, and advertisers/sponsors. By serving as the enclosure for these interactions, platforms can collect tolls (subscription fees or ad revenue) and, just as importantly, collect data.

⁸ Anne Helmond, “The Platformization of the Web: Making Web Data Platform Ready,” *Social Media + Society* 1, no. 2 (2015).

A defining trait of platforms is that they transform activities into data via what's known as datafication, rendering of interactions into quantifiable data points.⁹ Every song streamed, every video searched for, every second watched, every "like" or comment is logged. Data is then leveraged in service of the business model, whether through targeted advertising, personalized content recommendations to drive more usage, or even decisions about content production.

The major streaming platforms illustrate the nuances of the platform business model while all following its core logic. Netflix, for instance, runs on a subscription model where users pay a monthly fee for unlimited access to a catalog of films and TV shows. On the surface, this might seem like a straightforward digital update of the cable TV or video rental model. But Netflix's operation as a platform is far more complex. It heavily employs big data analytics to inform both its recommendations and its strategic decisions.¹⁰ By analyzing the viewing habits of its more than two hundred million subscribers, including granular data like when viewers pause, rewind, or abandon content, Netflix gains the ability and knowledge to tailor the user experience to individual viewers, driving engagement in a way that only a platform can enable. This data-driven personalization is a hallmark of platform logic. In effect, the service improves as users generate more and more data, which is then compared against, in the most basic form, likes and dislikes to generate similarities between users, leading to network effects where a larger user base yields better recommendations, attracting even more users, and the process repeats. In essence, Netflix's competitive edge comes not just from having content, but from having an architecture that mediates between content and viewers in a data-centric way.

YouTube, owned by Google, operates on a predominantly advertising-supported model and epitomizes the multi-sided platform dynamic. It attracts content creators who upload videos in hopes of reaching an audience and possibly earning a share of ad revenue; it attracts viewers who consume free content; and it attracts advertisers who pay to show ads to those viewers. YouTube's role is to intermediate all these parties on its technical platform by hosting videos, streaming them efficiently to users, and integrating targeted advertising.

⁹ José van Dijck et al., *The Platform Society: Public Values in a Connective World* (Oxford University Press, 2018).

¹⁰ Danial Amin, "How Netflix Used Data to Create House of Cards: A Revolutionary Approach to Content Creation," *Medium*, March 6, 2023.

Spotify provides another variant of the streaming platform model in the music domain. Spotify offers both a free, ad-supported tier and a premium subscription tier, effectively balancing two revenue models. As a platform, it interfaces between music listeners, music rights holders (record labels, independent artists), advertisers, and an ecosystem of third-party developers who can integrate with Spotify's API for things like playlist sharing or music analytics apps. Spotify's value proposition relies heavily on personalization and convenient, on-demand access to a virtually unlimited library of songs. It pioneered popular algorithmic playlists such as "Discover Weekly" and "Daily Mix," which analyze a user's listening history and broader trends to serve up custom-curated music playlists. The success of "Discover Weekly," which reportedly significantly increased user listening time,¹¹ underscores how data-driven selection is central to the platform's appeal. This mechanism is classified as "selection," whereby the platform drives the user by providing carefully curated content.¹² In other words, rather than the user actively choosing every song, the platform pre-selects and guides the user's choices, based on what the data suggests the user will enjoy. Ostensibly, this enhances user experience but also serves Spotify's business by keeping users engaged longer, thus exposing them to more ads or extending the life of their subscription. Like other platforms, Spotify also treats its aggregated user data as an asset—providing their insights to record labels and artists about listening patterns and even using data to negotiate power within the music industry.¹³

The platform model also extends to short-form video streaming such as TikTok. TikTok is quintessentially mobile-first and uses an algorithmically-curated feed known as the For You Page (FYP) as its core feature. TikTok's business model mixes advertising, sponsorships, and e-commerce integrations, but the user experience is entirely driven by continuous scrolling through an endless stream of personalized videos and advertisements. What sets TikTok apart is the remarkable degree of real-time data-driven adaptation. The platform's algorithm rapidly learns from each swipe, pause, or re-watch, tailoring the next pieces of content to maximize user retention. This creates a feedback loop where the platform's AI effectively curates a unique channel for each user. From a business perspective, this sticky engagement trans-

¹¹ Nils Aoun et al., "Discover Weekly: How the Music Platform Spotify Collects and Uses Your Data," *Montreal AI Ethics Institute*, May 26, 2022.

¹² van Dijck et al., *The Platform Society*, chap. 1.

¹³ Katie Shonk, "Streaming Toward Win-Win Negotiation: Spotify Upgrades Its Dispute Resolution with Taylor Swift," *Harvard Law School*, April 22, 2025.

lates into massive ad inventory and the potential for viral trends that can be monetized: for example, hashtag trends and challenges that advertisers co-opt into their marketing. TikTok embodies an automation of watching, in which users are “leaning back” and letting the algorithm feed them, while the platform meticulously monitors every interaction in order to refine the feed further.¹⁴ The most dystopian critiques of this kind of data-driven personalization highlight how these platforms create a situation where “we’re little lab rats in th[e] online forum,” continually experimented on by algorithms that test which content keeps us hooked.¹⁵ The platform’s goal is deeply aligned with the infrastructure of data collection: More data yields better personalization, which yields more engagement, which yields more data, in a continuous cycle.

One cannot overlook Twitch, the Amazon-owned, ostensibly video gaming-centric live streaming platform. Twitch primarily monetizes through a combination of advertising and viewer subscriptions/donations to individual streamers, of which Twitch takes a cut. As a live streaming platform, its emphasis is on real-time community and interaction, such as live chat, emotes, donations, etc. This fosters a different dynamic from Netflix or TikTok that attempts to optimize content delivery to individuals. Twitch, on the other hand, has built an architecture that supports vibrant communal experiences at scale. In many cases this has paid off, as popular streamers might have tens of thousands of viewers chatting simultaneously, yet it is important to note that the vast majority of Twitch streamers (93.3%) have five viewers or less.¹⁶ This illuminates the allure of both the community-driven experience and of streaming’s potential (albeit very unlikely) to become a full-time job.

The technical challenges here become part of the platform’s competitive advantage: Ultra-low latency video delivery and synchronized chat are infrastructure feats that Twitch continuously works to optimize. From a business standpoint, Twitch’s data collection is not about recommending on-demand content, since live content is ephemeral, but it still gathers extensive metrics on viewership patterns, chat sentiment, etc., which can inform which streamers get promoted or which events are spotlighted. Indeed, there is a recommendation system similar to that of other stream-

¹⁴ Lisa Parks et al., eds., *Media Backends: Digital Infrastructures and Sociotechnical Relations* (University of Illinois Press, 2023).

¹⁵ Sarah Childress, “Mark Andrejevic: We Are All ‘Lab Rats’ Online,” *Frontline*, PBS, February 18, 2014.

¹⁶ Bill Cooney, “New Twitch Stats Reveal How Few Viewers Are Needed to Be a ‘Top’ Streamer,” *Dexerto*, March 6, 2021.

ing platforms; however, it is dependent upon channels that are, at that very moment, live. Moreover, Twitch demonstrates the diversity of content domains that streaming platforms can capitalize on. Streaming is not just pre-recorded media like songs, shows, and movies, but human activities, such as video gaming and performances in real-time.

While each platform has its own flavor, they share common strategies typical of platform capitalism: user acquisition through network effects, monetization through data-driven decision making, data-driven advertising or subscription at scale, and an ecosystem of control via APIs and platform governance. Three relevant types of platforms typify mobile streaming: advertising platforms (e.g., Google/YouTube, Facebook), cloud platforms (e.g., Amazon Web Services, which underlies many streaming services), and product platforms (e.g., eBay, Amazon).¹⁷ Many streaming services combine and employ various aspects of these or converge them into one ecosystem.

YouTube is an advertising platform; Netflix could be seen as a kind of content platform with a subscription base; Spotify is a hybrid; and Amazon is a special case that spans multiple platforms in that it runs both Twitch and AWS infrastructures, and Prime Video as a streaming service integrated into its e-commerce ecosystem. Across all these platforms, user data collected via mobile—location, app usage, listening/viewing habits, and interaction patterns are central. The mobile interface becomes a key site of data extraction, encouraging platforms to expand into more corners of user life. YouTube Shorts, Spotify’s podcast autoplay, and Netflix’s mobile games are all strategic moves to keep users within the app and to collect richer behavioral data. These examples show that “streaming” isn’t a single industry silo, but a cross-industry model that integrates entertainment, tech, and telecommunications into one ecosystem of on-demand content.

This logic often undermines user privacy and public interest. Mobile design features like infinite scroll, autoplay, and notification nudges are optimized to maximize engagement, but raise ethical concerns. A few dominant mobile platforms now control the bulk of mobile cultural consumption and communication. Their algorithms not only shape what is recommended, but also which creators succeed, which genres thrive, and which voices are heard. The result is a winner-take-all mobile ecosystem, where YouTube dominates mobile video, Spotify leads in music streaming, and Netflix holds firm in subscription-based entertainment. This consolidation hands enormous power to a few companies to shape cultural consumption and user data

¹⁷ Srnicek, *Platform Capitalism*, 49.

flows. Even well-resourced newcomers like Disney+ struggle to disrupt these entrenched mobile behaviors. In effect, the mobile streaming environment is consolidated, data-driven, and platform-governed—an infrastructure as central to contemporary media as broadcast towers and cable grids. In the next section, we turn to those underlying infrastructures—the wireless networks and mobile systems—that materially enable these platform-driven, personalized media experiences.

Wireless Networks and Mobile Infrastructure

We often don't think about mobile streaming via videocalls, but this relatively recent development has major consequences for platforms and wireless service providers alike. For example, Apple's FaceTime feature was initially restricted to Wi-Fi connections, as AT&T's 3G network couldn't support the bandwidth demands of video calls. This limitation underscored the inadequacies of existing wireless infrastructure in handling emerging streaming technologies. The challenges were further highlighted during Apple's 2010 Worldwide Developers Conference (WWDC), where Steve Jobs attempted to demonstrate FaceTime. The presentation faced significant Wi-Fi interference, causing delays and emphasizing the need for more robust wireless networks to support seamless streaming experiences. These events exemplify how the evolution of wireless infrastructure has been crucial in enabling the mobile streaming capabilities we often take for granted today.

If digital platforms provide the portals and economic engines for mobile streaming, wireless communications infrastructure provides the roads and highways those portals depend on. The notion of architectures of mobility directs our attention to the physical and code-as-protocol layers that enable users to access streaming media while on the move—principally, cellular and Wi-Fi networks that often undergird mobile connectivity. The evolution of cellular technology from 3G to 4G and now 5G has been a fundamental enabler of the mobile streaming boom. The value of mobile networks lies in providing ubiquitous connectivity, an “always-on” data stream wherever a user goes. This is the essence of mobile personalization—creating an environment in which mobile streaming media practices are not tied to a fixed place. Mark Andrejevic evocatively described the result as “a ubiquitous electromagnetic atmosphere—not merely extension across space, but a ‘filling-in’ of space, so that wherever one goes, the network can be accessed.”¹⁸ In practical terms,

¹⁸ Mark Andrejevic, “Media and Mobility,” in *The International Encyclopedia of Media Stud-*

this ubiquitous connectivity is achieved by the dense infrastructure of cell towers, underwater cables, data centers, exchange points, routers, and our own devices that blanket populated areas with wireless internet access.

The progression from 3G to 4G to 5G illustrates how each generation of wireless infrastructure opened new possibilities for streaming media. 3G networks (starting early 2000s) were the first to support reasonable data throughput on phones (~hundreds of kilobits per second up to a few megabits per second later in the life cycle). With 3G, one could do basic web browsing and perhaps stream highly compressed audio or low-resolution video, such as YouTube's mobile site in its early days. Yet 3G was often barely sufficient for smooth video and was limited by high latency. The rollout of 4G LTE (Long Term Evolution) around 2010 was transformative. Mobile bandwidth became comparable to home broadband, reaching tens of megabytes per second, with lower latency. This allowed users to stream 720p or, depending on their area of coverage, 1080p HD video on a phone or tablet via cellular, which had been impractical before. Telecommunications companies in Western markets experienced surging data traffic on their 4G networks largely due to video streaming, and by the second half of the 2010s, video dominated mobile data. One industry report noted that in North America, Netflix and YouTube alone accounted for about half of downstream mobile internet traffic by 2019.¹⁹

The network infrastructure had to keep pace. Carriers invested in upgrading backhaul (the connections from cell towers to the internet backbone), deploying more cell sites to increase capacity, and implementing technologies like LTE-Advanced with carrier aggregation to boost speeds. They also had to adjust data pricing models; many moved from unlimited plans to tiered data plans in the early 2010s to manage the explosion in use. Competitive pressures later brought back more "unlimited" offerings, though this often came with throttling caveats for heavy use or lower quality video streaming by default.

5G, the fifth-generation standard, began deployment in the late 2010s and is still being rolled out across the US. While 5G's real-world impact is still unfolding, it promises significantly higher speeds, in the gigabit per second range under ideal conditions, and, importantly for new media applications, much lower latency of 10 milliseconds or less in optimized scenarios. The

ies, ed. Angharad N. Valdivia, vol. 1, *Media History and the Foundations of Media Studies*, ed. John Nerone (Blackwell, 2013), 521.

¹⁹ Troy Dreier, "Video Will Be 78% of Mobile Data Traffic by 2021, Says Cisco," *Streaming Media*, February 7, 2017.

arrival of 5G has been heralded as enabling next-generation streaming experiences: ultra-high-definition 4K or 8K video on mobile, flawless VR/AR streaming, and cloud gaming with latency nearly as low as a local machine.²⁰ Early reports indicate that 5G networks “ensure flawless synchronization,” making it ideal for live sports, remote interviews, and virtual production.²¹ In essence, 5G is designed to make the mobile network feel “invisible”—so fast and responsive that users cannot distinguish between wired and wireless connectivity.

While 5G coverage and device uptake will take several years to mature, many streaming platforms have prepared by supporting higher quality streams and new features for 5G users. For example, Netflix and YouTube support HDR and 4K on mobile where the network allows, and cloud gaming services offered by Xbox, Steam, and PlayStation are poised to benefit from 5G’s low latency. Telecommunications providers often market 5G by highlighting streaming capabilities. Verizon and AT&T adverts show users seamlessly live-streaming or watching HD content on the go.

Control over these networks confers power, which is evident in debates over net neutrality and incidents like the 2014 Netflix–ISP peering dispute, where infrastructure owners could leverage quality of service to negotiate with content providers. Particularly in the United States and Europe, the principle of “net neutrality”—that internet service providers (ISPs) should treat all data equally without undue discrimination—has been hotly debated,²² ISPs have sometimes chafed at services like Netflix or YouTube consuming massive bandwidth on their networks without direct compensation. Conversely, platform companies argue that users have already paid the ISP for internet access and should get the content they request unimpeded.²³ This tension led to incidents like the 2014 peering dispute where Netflix traffic to Verizon and Comcast customers slowed down until Netflix agreed to pay for direct connections.²⁴

The debate and influence of those who regulate streaming environments can shape mobile streaming architectures. For example, zero-rating

²⁰ “Everything You Need to Know About 5G,” *What Is 5G?*, Qualcomm, accessed May 12, 2025.

²¹ “How 5G Is Transforming Live Broadcasting and Remote Production,” *Medialooks*, September 2024.

²² Rahul Awati and Kate Gerwig, “Net Neutrality,” *TechTarget*, May 2, 2024.

²³ Eriq Gardner, “Netflix Supports Net Neutrality Rules in Fight Against ISPs,” *Hollywood Reporter*, April 25, 2023.

²⁴ Nicholas Cartier, “Netflix’s System Architecture Represents a Rare Competitive Edge,” *Streaming Media*, October 24, 2023.

practices, where certain streaming services don't count against mobile data caps, have been used by carriers as a competitive tool. T-Mobile's "Binge On" program allowed customers in the US to stream from certain video services without using data, albeit at DVD quality, which arguably favored established partners and raised net neutrality questions. In the EU, some regulators cracked down on or scrutinized such practices to ensure a level playing field for content providers. These policies effectively determine whether telecom infrastructure remains a neutral conduit or whether it can act like a platform gatekeeper itself, prioritizing some content. The general trend in Western markets has been toward neutrality (especially in the EU), but with ongoing battles, particularly in the US, neutrality rules have waned with different administrations. For our purposes, the key point is that the design of mobile infrastructure governance directly affects streaming platforms: If an ISP throttles video or exempts one service from caps, it shapes user behavior and the competitive landscape of platforms.

Another aspect of wireless infrastructure is the integration of Wi-Fi and fixed networks into the mobile streaming experience. A large portion of "mobile" streaming occurs on Wi-Fi rather than via cellular data, since Wi-Fi is often less capped by connection speeds and faster. This has led to mobile apps typically allowing users to choose video quality based on network (lower on cellular, higher on Wi-Fi by default) to manage data use. Also, ISPs increasingly offer seamless bundling between cellular and Wi-Fi networks. For example, a smartphone may automatically switch to known Wi-Fi networks to offload traffic from cellular networks. Some cable companies in the US (like Comcast) leveraged their extensive Wi-Fi hotspot networks to create quasi-mobile networks, allowing their customers to stream outside the home without being on a traditional cellular plan. This convergence of Wi-Fi and cellular networks illustrates how "mobile infrastructure" is a patchwork of different network types. Nonetheless, from the user perspective, what matters is that a connection is available—whether via 4G, 5G, or Wi-Fi—to enable streaming. Castells's notion that being connected anywhere is the new normal of communication holds true, as people expect uninterrupted access to media as they move through different spaces.²⁵

During peak events, such as a major sports championship being streamed or a new Netflix series release, cellular networks may experience localized spikes in demand. Carriers deploy traffic management techniques such as quality-of-service (QoS) prioritization, compression, and sometimes throt-

²⁵ Castells, "Communication, Power and Counter-Power," 238.

ting heavy users to maintain overall network performance.²⁶ The back-end systems of telecommunication networks, like content-aware routers or caching within ISPs, also play a role. Many ISPs and mobile carriers deploy caches for popular streaming platforms—often as part of the platform’s own CDN program (e.g., Netflix Open Connect appliances or Google Global Cache for YouTube). These caches are infrastructure nodes that store copies of popular content within an ISP’s network, so that if multiple users in that region request the same video, it can be delivered from the local ISP cache rather than each request traversing the wider internet.²⁷ This significantly reduces bandwidth costs and improves latency. Such arrangements again blur the lines of platform and infrastructure. The platform (Netflix/Google) provides the caching servers and content, while the ISP provides power, housing, and connectivity for them, each benefiting from more efficient delivery. The architectural design here is collaborative—a layered system optimized for streaming quality.

Spectrum allocation and cell tower siting, while highly technical, are also part of the narrative. Spectrum allocation is the process of assigning specific frequency ranges to different types of wireless communication services (like mobile phones, radio, or TV) to avoid interference and ensure efficient use of the airwaves. Cell tower siting, on the other hand, is the process of selecting and approving locations for building cell towers, considering factors like signal coverage, zoning laws, and community impact. Mobile streaming’s success made video a must-consider traffic type when planning networks. 5G in the form of high-frequency millimeter wave bands offers huge capacity but short range, meaning it can support many 4K streams in a stadium or downtown (spaces of dense usage) but requires many small cells. Lower frequency bands cover wider areas but are more easily congested with video load. Carriers are spending billions at spectrum auctions (the US auctioned mid-band C-Band for 5G in 2021 for over \$80 billion,²⁸ for instance) to secure the frequencies needed to carry future streaming-intensive traffic. In a sense, streaming media has become a driver of infrastructure investment cycles.

Wireless infrastructure provides the pervasive connectivity that allows streaming platforms to reach users while they are on the go, and the continuous upgrades and policies of these networks enable mobile streaming’s

²⁶ Samir Marwaha, “Sandvine’s 2024 Global Internet Phenomena Report: Global Internet Usage Continues to Grow,” AppLogic Networks (blog), April 10, 2024.

²⁷ “What Is Caching?” *Fortinet*, accessed May 12, 2025.

²⁸ “Auction 107: 3.7 GHz Service,” Federal Communications Commission, accessed May 12, 2025.

growth. The relationship is symbiotic. User demand for mobile streaming drives network investment, and improved networks drive further innovative uses of streaming. Effectively, a network has been built where the movement of messages (media streams) are tracked alongside the movement of people. Smartphones constantly communicate with cell towers, negotiating bandwidth for the next segment of a video or song, creating a seamless illusion that media resides in the “cloud.” This always-on connectivity has effectively merged with our daily routines. Media consumption is no longer an activity bound to a living room or a wired desktop, but something that can occur in the car, on a walk, in a café, or anywhere we carry a device.

Taken together, the wireless infrastructure component of architectures of mobility underpins everything the platforms do. It is the less visible part of the system (users notice Netflix or YouTube, not TCP/IP packets, cell handovers, data centers, etc.), yet it is decisive in shaping what is possible. A limitation in infrastructure (e.g., data caps, congested networks, down towers or cables) directly limits the platform’s usage and revenue; conversely, advancements such as 5G open new strategic opportunities for platforms to offer varied types of media, including higher resolution content and interactive experiences. The next section examines how platforms capitalize on that connectivity—through data and algorithms—to personalize content and manage the constant stream of information.

Big Data, Algorithms, and APIs: The Backend of Mobile Streaming

We now turn our attention to the less-visible backend of mobile streaming: the big data pipelines, algorithmic systems, and APIs that power personalized content delivery. As of 2024, there are over five hundred hours of video uploaded to YouTube every minute, and the platform has over two billion logged-in monthly users worldwide.²⁹ To manage this deluge of content and connect viewers with videos that will keep them engaged, YouTube relies on algorithmic curation. Its recommendation algorithm and “Up Next” autoplay feature drive a large portion of views, learning from each user’s history and viewing/interaction patterns across similar users. Platforms like YouTube and Facebook do not merely mediate public discourse, “they constitute it”

²⁹ Robin Geuens, “How Many Hours of Video Are Uploaded to YouTube Every Minute?” SOAX, October 30, 2024.

through the design of their algorithms and interfaces.³⁰ By deciding what content is prominently recommended or filtered out, the platform architecture shapes audience tastes and trends and responds accordingly. Work on the “politics of platforms” emphasizes that companies like YouTube strategically frame themselves as neutral conduits or open marketplaces, yet in practice they continuously intervene, through moderation policies, recommendation systems, and interface nudges to organize content and user behavior in profitable ways.³¹ As such, YouTube’s business model is inseparable from its technical system of data collection and algorithmic sorting, which maximizes advertising opportunities (more watch time equals more ads viewed) but also raises concerns about filter bubbles and the promotion of sensational or extremist content as a side effect of engagement-driven algorithms. These hidden algorithms exert power by governing what audiences see (and don’t see), effectively impacting major aspects of public discourse.

Mobile streaming platforms are often praised for their, ostensibly, slick interfaces and vast content libraries, but their true power resides in the less-visible backend systems: the big data repositories, algorithmic recommendation engines, and APIs that together orchestrate what content is delivered to whom. These elements form the intelligent layer that sits atop the physical infrastructure. They enable personalization at scale, real-time interactivity, and the integration of third-party services, all of which are crucial to the modern streaming experience. This section examines how big data and algorithms operate within mobile streaming architectures, and how application programming interfaces function as both technical and governance mechanisms in the platform ecosystem.

Data collection is a core mechanism of platform ecosystems; it is the ongoing effort to convert every aspect of user activity into quantifiable data.³² Mobile streaming intensifies this process, as smartphones function not only as media devices but also as personal sensors. They capture contextual information such as location, device type, and social interactions in addition to content choices. Each time a user taps on a video or song, the system logs what was selected, when it was accessed, possibly where the user was, and whether the content was completed. Aggregated across millions of users, this generates big data characterized by high volume, speed, and variety. These data streams are typically funneled into cloud databases

³⁰ Tarleton Gillespie, *Custodians of the Internet: Platforms, Content Moderation, and the Hidden Decisions That Shape Social Media* (Yale University Press, 2018).

³¹ Tarleton Gillespie, “The Politics of ‘Platforms,’” *New Media & Society* 12, no. 3 (2010).

³² van Dijck et al., *The Platform Society*, 101.

and data lakes maintained by the platforms. YouTube tracks detailed viewing and search histories, Spotify logs second-by-second listening behavior, and Netflix monitors viewing patterns across time and demographics. This practice of datafication forms the basis of the “digital enclosure,” where user activity is continuously monitored and mined for informational value.³³ In the context of streaming, the enclosure is both literal within the confines of the app where every action is tracked, and metaphorical in reflecting the broader condition of having to accept dataveillance in exchange for access to services.

The implications of pervasive data collection are multifaceted. On the one hand, it enables unprecedented mobile personalization and convenience. When Netflix recommends a show that resonates with a user, or when Spotify surfaces a new favorite song, those moments of discovery are enabled by the analysis of vast data sets. However, this personalization comes with a tradeoff. Platforms hold immense knowledge about users, while users themselves have little insight into what has been collected or how it is interpreted. This disparity is often called the big data divide, highlighting how platforms gain strategic advantage through opaque data practices.³⁴ Users only see the output, such as a recommendation or playlist, but not the layers of surveillance and inference behind it. This data is often used for purposes beyond personalization, including targeted advertising, dynamic pricing, and the sale of anonymized trend data. Andrejevic warns that we now exist in a system of continuous observation where individuals effectively serve as experimental subjects, contributing unknowingly to A/B tests and machine learning models that refine platform behavior in real time.³⁵ This situation illustrates a power asymmetry: Platforms wield significant informational power over users, who remain largely in the dark about how their data is used. Streaming services regularly experiment with interface changes, algorithm adjustments, and content placement to see what drives engagement. Netflix, for example, tests thumbnail images and descriptions, while Facebook famously altered users’ news feeds in a study on emotional contagion.³⁶ These examples illustrate how platform governance increasingly relies on iterative, data-driven experimentation.

³³ Mark Andrejevic, “Surveillance in the Digital Enclosure,” *The Communication Review* 10, no. 4 (2007).

³⁴ Mark Andrejevic, *Infoglut: How Too Much Information Is Changing the Way We Think and Know* (Routledge, 2013).

³⁵ Parks et al., *Media Backends*, 13.

³⁶ Adam D. I. Kramer et al., “Experimental Evidence of Massive-Scale Emotional Conta-

At the heart of this experimentation by major streaming platforms is an algorithmic curation system that determines what content is shown or suggested to users. These algorithms are the computational manifestation of the platform's editorial logic. In earlier eras, media consumption was largely scheduled and curated by human editors or programmers (TV channel line-ups, radio DJs, etc.), but in the streaming era this role has increasingly been assumed by algorithms processing big data. Netflix's homepage is not a static catalog, but a personalized montage of rows and thumbnails uniquely generated for each viewer. YouTube and TikTok feeds are virtually endless, generated on the fly based on one's past behavior and real-time feedback. Spotify's playlists like "Discover Weekly" are assembled by algorithms that analyze millions of tracks and listening profiles. In short, algorithmic personalization is taking place at a mobile scale.

These recommendation engines typically combine several techniques. Collaborative filtering identifies patterns among users with similar tastes. Content-based filtering matches characteristics of previously liked content to suggest similar options. More recently, deep learning models have been introduced to detect subtle and complex correlations in user behavior.³⁷ YouTube's algorithm considers hundreds of signals, including video topic, watch time, click-through rate, and recency, to rank videos in the "Up Next" queue. TikTok also analyzes micro-behaviors such as how long a user watches a video, whether they replay it, or how quickly they scroll past. While these systems are designed to improve user satisfaction by surfacing relevant content, their deeper goal is to maximize engagement to support the business model. But remember, platforms do not simply mediate public discourse; they help constitute it.³⁸ Recommendation algorithms and interface features influence what people see and, in turn, what kinds of media rise to cultural prominence. For example, if Spotify's algorithm favors songs that are played to the end, artists may adjust their music to minimize listener drop-off. If YouTube rewards longer watch times, creators may extend video length or design more compelling openings. These algorithms quietly influence both user choices and the direction of creative production.

gion Through Social Networks," *Proceedings of the National Academy of Sciences of the United States of America* 111, no. 24 (2014).

³⁷ Paul Covington et al., "Deep Neural Networks for YouTube Recommendations," in *Proceedings of the 10th ACM Conference on Recommender Systems* (Association for Computing Machinery, 2016).

³⁸ Tarleton Gillespie, "Platforms Are Not Intermediaries," *Georgetown Law Technology Review* 2, no. 2 (2018).

While algorithmic curation can enhance usability by helping users navigate an overwhelming amount of content, it has also sparked considerable criticism. One prominent concern is the loop of content in which users become trapped, where algorithms become so effective at predicting preferences that they mostly deliver content aligned with a user's existing tastes or beliefs, thus creating filter bubbles and/or echo chambers. This limits exposure to differing viewpoints or unfamiliar material. The Proceedings of the National Academy of Sciences points out that YouTube's algorithm has frequently been accused of contributing to filter bubbles and guiding users toward increasingly radical content.³⁹ In the political sphere, critics argue that the recommendation engine can steer users toward more extreme or conspiratorial videos in its effort to maximize watch time. Even beyond political content, recommendation systems can narrow the range of media a user encounters. A listener who is repeatedly served the same style of music, or a viewer shown only one genre of television, may find their media habits becoming increasingly uniform. While platforms like Netflix and YouTube sometimes introduce unexpected or trending content into a user's feed to encourage exploration, personalized recommendations still dominate and largely shape the user's experience.

There are also concerns about fairness and transparency. Algorithms can unintentionally amplify certain types of content or creators while marginalizing others. For instance, if user behavior data suggests that one demographic generates higher engagement, the system may end up recommending content featuring that group more frequently, reducing visibility for others. Yet these decisions are largely invisible to users and creators because the algorithms are proprietary and in constant flux. Platforms treat them as trade secrets, offering only vague explanations through creator guidelines or occasional blog updates. This lack of transparency has led to growing calls for algorithmic accountability, especially as these systems shape the distribution and visibility of culture online.⁴⁰

In addition to shaping recommendations, algorithms also govern moderation and platform health. YouTube's Content ID system scans uploaded content for copyrighted material and can automatically block or demonetize

³⁹ Muhammad Haroon et al., "Auditing YouTube's Recommendation System for Ideologically Congenial, Extreme, and Problematic Recommendations," *Proceedings of the National Academy of Sciences of the United States of America* 120, no. 50 (2023).

⁴⁰ Frank Pasquale, *The Black Box Society: The Secret Algorithms That Control Money and Information* (Harvard University Press, 2015); Safiya Umoja Noble, *Algorithms of Oppression: How Search Engines Reinforce Racism* (NYU Press, 2018).

content on behalf of rights holders. Facebook and Twitch rely on automated systems to detect and flag potentially harmful material, such as hate speech or nudity, which is then reviewed by human moderators. These tools have a direct impact on what content remains available, affecting the cultural landscape users interact with. Indeed, moderation is a foundational function of platforms that is deeply tied to the values and business models of the platform owners. On TikTok, this becomes especially apparent as the algorithm promotes engaging videos while also suppressing content flagged as violating community guidelines. In some cases, TikTok has been accused of limiting the reach of political content or content from marginalized creators, raising further concerns about bias and suppression.⁴¹ Ultimately, algorithmic curation is not just about improving user experience, but it also reflects legal, commercial, and ethical decisions embedded in platform design.

Algorithmic curation is a defining feature of mobile streaming platforms. It enables personalization at scale, something no team of human editors could achieve for billions of users. However, it also introduces an automated, data-driven layer of governance over media consumption that is closely tied to corporate goals. These algorithmic systems are part of the broader architectures of mobility, not only through physical infrastructures like networks and devices, but also through the software and logic that govern information flow. Just as road design affects traffic patterns and accessibility, algorithm design shapes the visibility of content and influences what users see. Recognizing this influence is central to media literacy. Users must understand that what appears on their screens is not neutral, but the outcome of deliberate algorithmic processes. In response to growing concerns, scholars and policymakers have proposed interventions such as transparency reports and algorithmic audits. In the European Union, the Digital Services Act mandates that major platforms provide external researchers with access to their algorithms for the purpose of identifying systemic risks.⁴² These regulatory moves suggest a shift in how algorithms are viewed—not just as technical tools, but as components of public infrastructure that warrant oversight and accountability.

⁴¹ Robyn Caplan and Tarleton Gillespie, “Tiered Governance and Demonetization: The Shifting Terms of Labor and Compensation in the Platform Economy,” *Social Media + Society* 6, no. 2 (April 2020).

⁴² European Union Regulation (EU) 2022/1925 of the European Parliament and of the Council of 14 September 2022 on Contestable and Fair Markets in the Digital Sector and Amending Directives (EU) 2019/1937 and (EU) 2020/1828 (Digital Markets Act), *Official Journal of the European Union* L265/1 (2022).

In sum, controlling the backend has become as crucial as owning the network or the content. This blur between running a platform and running infrastructure leads us to the final issue: the full convergence of platforms and infrastructures, and its consequences.

APIs and the Merging of Platforms with Infrastructure

While algorithms manage the flow of content within platforms, application programming interfaces (APIs) govern how platforms interact with the outside world and with other software. APIs are the formal channels through which external applications or devices request data or services from a platform. They are a foundational element of modern digital ecosystems, including mobile streaming. For example, when a smart TV or Chromecast plays a Netflix show, it relies on Netflix's API to access the catalog and deliver the content. Similarly, a third-party website embedding a YouTube video uses YouTube's embed API. Spotify offers APIs that allow developers to create applications capable of retrieving song data or controlling playback, provided the user grants permission. In mobile streaming, APIs enable essential integrations such as content sharing to social media, single sign-on through services like Facebook or Google, and data reporting for analytics and advertising. These functions illustrate how APIs extend the reach of streaming platforms across devices, apps, and services.

Beyond their technical role, APIs also work as tools of platform governance and strategy. Anne Helmond argues that APIs are not just interfaces for data access but also mechanisms of control, allowing platforms to shape the broader digital ecosystem.⁴³ By controlling what data or capabilities are accessible—and to whom—platforms can guide third-party development in ways that align with their business goals. Twitter offers a clear example. In its early years, the company maintained a relatively open API that encouraged a flourishing ecosystem of third-party clients, which helped grow the platform. However, as Twitter shifted toward monetization through advertising within its own interface, it began restricting API usage. By 2023, it had fully revoked access for many third-party apps, effectively shutting them down.⁴⁴ This shift illustrates how platforms use API governance not

⁴³ Anne Helmond, "The Platformization of the Web: Making Web Data Platform Ready," *Social Media + Society* 1, no. 2 (2015).

⁴⁴ Mitchell Clark, "The Third-Party Apps Twitter Just Killed Made the Site What It Is Today," *The Verge*, January 22, 2023.

only to facilitate innovation but also to consolidate control in alignment with commercial interests.

In the realm of streaming media, similar patterns of platform control and governance play out. During its expansion phase, Netflix offered a public API between 2008 and 2014 that allowed external applications to display titles, manage queues, and build Netflix-based tools.⁴⁵ However, once Netflix had achieved wide adoption and could ensure consistent access through its official apps on all major devices, it shut down the public API in 2014. This move signaled a shift toward full control of the user experience, managed entirely within its own ecosystem. YouTube, in contrast, still maintains a relatively open API. Despite its openness, YouTube imposes clear limitations, including caps on requests and strict terms that prohibit alternative user-facing designs. These rules ensure that while YouTube's reach extends across the web, it retains core control. Videos must still play in YouTube's player, advertisements are served by YouTube, and any API client that violates policy risks losing access.⁴⁶ Spotify offers another example. Its API supports integration with smart home devices, enabling native Spotify functionality on speakers and other systems. It also connects to services like Last.FM and other social media platforms to display currently playing tracks. These integrations make Spotify more embedded in everyday user routines. However, Spotify carefully governs what data is made accessible. For many years, it withheld access to individual user's listening histories, citing privacy concerns and perhaps a desire to avoid enabling competitors or third-party data mining.⁴⁷ These examples illustrate how APIs function as selective gateways, tools for expansion and entrenchment, but also instruments of strategic control.

From a systems architecture perspective, APIs make mobile streaming scalable and modular. Whether users access YouTube via a smartphone, smart TV, game console, or browser, these different devices typically connect to the same backend API endpoints to fetch content and log interactions. This centralization allows platforms to update recommendation systems or content libraries in one place. APIs also enable cross-platform integration that contributes to infrastructural convergence. For example, authentication systems such as Google or Facebook login are made possible by those com-

⁴⁵ Ryan Lawler, "Netflix Will Shut Down Public API Support For Third-Party Developers On November 14," *TechCrunch*, June 13, 2014.

⁴⁶ "YouTube Data API Overview," Google Developers, accessed May 12, 2025.

⁴⁷ Bartosz Janota and Robert Stephenson, "Spotify's Event Delivery: Life in the Cloud," *Engineering* (blog), Spotify R&D, November 12, 2019.

panies' APIs. In streaming, ISP companies sometimes bundle subscriptions to services like Netflix or Disney+, with backend integration happening through APIs. In these cases, a carrier's set-top box may directly pull streams from the platform, creating a seamless experience that blurs the boundary between internet service provider infrastructure and the streaming service itself. For the end user, it often appears as one unified system.

Controlling the API also means controlling the terms under which others can interact with a platform's data or features. This control gives platforms significant leverage in shaping how their ecosystems function. This is accomplished when platforms increasingly assume infrastructural roles, as they can then impose their own technical and policy standards on third parties.⁴⁸ A clear example is seen in the mobile ecosystems governed by Google and Apple. Both companies provide APIs within Android and iOS that manage essential functions such as media playback, notifications, and in-app purchases. Streaming services must rely on these APIs to operate correctly on mobile devices, and in doing so, they often must comply with platform-specific rules.

For instance, Apple requires the use of its in-app payment system and enforces restrictions on alternative payment methods. It also mandates compatibility with specific video codecs supported by its operating system. In this way, the API becomes a key site where the platform power of app stores intersects with that of streaming services. The conflict between Epic Games and Apple offers a striking example. When Epic attempted to bypass Apple's in-app purchase API in the mobile version of Fortnite, Apple removed the game from its App Store. This response demonstrates the rigidity of these technical and economic architectures and highlights how tightly control over APIs is linked to broader questions of platform governance and market power.

The merging of platform and infrastructure is clearly visible in the cloud computing services that many streaming platforms depend on. Companies like Amazon Web Services (AWS), Microsoft Azure, and Google Cloud provide infrastructure-as-a-service through APIs. These services allow platforms to store data, rent servers, and use content delivery networks as needed. Netflix was one of AWS's most prominent customers for cloud computing, even as it developed its own content delivery network, Open Connect. Spotify similarly relies on Google Cloud for its backend infrastructure. In these arrangements, a platform such as Netflix or Spotify is built on top of another

⁴⁸ Plantin et al., "Infrastructure Studies Meet Platform Studies," 307.

platform's infrastructure, interacting entirely through API calls. This kind of layered dependence illustrates how some platforms effectively become the infrastructure for others.

The internet now functions through these kinds of layers. Authentication systems, mapping services, payment gateways, and analytics platforms all rely on APIs, meaning that much of an app's functionality may come from external services. The system is interoperable in theory, as any service can connect with another if APIs are available. For instance, a developer could create an app that queues a YouTube video using a voice command by combining YouTube's API with a voice recognition API. However, the system is also tightly controlled. Each API is owned and governed by a platform that sets specific usage terms, rate limits, and permissions.

The control of APIs is yet another consequence of platforms as infrastructures via the expansion of platforms into various sectors through APIs and modular integration. In mobile streaming, this is especially evident. News outlets distribute video through Twitter and Facebook APIs, sports apps incorporate YouTube highlights, and music platforms allow users to share songs directly to social media. The architecture of mobility now consists of a dense mesh of API connections layered on top of physical infrastructure. This architecture can be visualized as a stack: Physical infrastructure like fiber optics and cell towers forms the base; above that is cloud infrastructure and content delivery networks accessed through internet protocols; platform logic and data exist on the next layer and are accessed through APIs; and the top layer consists of the user-facing apps. Each layer communicates with the next through defined protocols, such as HTTP and JSON-based APIs.

By managing both algorithms and APIs, large streaming platforms control not only the internal user experience but also how their services connect with the broader digital ecosystem. Amazon is a prime example of this convergence. It owns Twitch as a content platform, operates AWS as a cloud infrastructure provider, sells hardware such as phones, tablets, TVs, etc., and runs the Prime Video service. This integration allows Amazon to coordinate across layers, giving priority to its own services on its hardware or using AWS data to enhance performance across its platforms. Google has a similar structure with the Android operating system (OS): as a content and algorithm-driven platform, with Google Cloud for infrastructure. This vertical integration has raised antitrust concerns,⁴⁹ but it also makes these platforms extremely resilient and difficult to challenge. A startup may create

⁴⁹ Lina M. Khan, "Amazon's Antitrust Paradox," *The Yale Law Journal* 126, no. 3 (2017).

an innovative streaming app, but without the economies of scale provided by major cloud providers or access to user data for recommendation systems, it faces significant structural barriers. Unless of course, the start-up is purchased by one of the conglomerates, which is sometimes the goal to begin with.

Platforms as Infrastructures: Convergence and Consequences

If there's one continual refrain this chapter, it's that the line between platforms and infrastructures has become increasingly difficult to define, and this convergence has profound consequences for mobile personalization and power. The technologies and business arrangements that allow a user to stream a movie on a smartphone while commuting reflect a convergence of roles that were once, ostensibly, separate. In the past, traditional infrastructures such as telephone networks or cable television systems functioned as carriers, typically seen to only transmit content from distinct service providers. While power relations in these delivery mechanisms certainly existed in earlier forms of mass media, these pale in comparison to the tangled web of platforms, wireless infrastructures, data centers, APIs, and proprietary algorithms of contemporary streaming. Today, companies like Google, Amazon, and Netflix operate not only the platforms that deliver content and services but also the infrastructure that supports global communication. These companies maintain data centers, undersea cables, and foundational services that are now integral to digital life.

Mobile streaming offers concrete examples of this convergence. Google functions simultaneously as a platform and an infrastructure provider. Through YouTube, it connects video creators and viewers, while Android supports app development and mobile services. At the same time, Google owns and operates undersea fiber-optic cables and one of the world's largest content delivery systems. It is important to note that the physical infrastructure, particularly undersea cables, was once maintained by a consortium of telecommunication companies; however, this infrastructure is increasingly privatized.⁵⁰ Android itself serves as a mobile infrastructure that other applications depend on to function. Similarly, Amazon operates Amazon Web Services (AWS), which powers not only its own streaming offerings but also those of direct competitors. Netflix, for example, has long relied on AWS for cloud services, effectively paying Amazon to host its content infrastructure

⁵⁰ Phil Gervasi, "Diving Deep into Submarine Cables: The Undersea Lifelines of Internet Connectivity," *Kentik* (blog), March 28, 2023.

while competing with Amazon Prime Video for market share. This layered interdependence reflects a defining feature of the current digital ecosystem. Companies that compete in the consumer marketplace may simultaneously rely on one another behind the scenes. The result is a complex and often contradictory system, where infrastructure and platform roles are deeply entangled, and where competitors can also be business partners due to the way modern digital services are built and delivered. In effect, streaming's infrastructural logic knits together what were once distinct industries (tech, telecom, and media) into a single, vertically integrated system.

Another facet of convergence is when infrastructure owners expand into providing content services directly. Telecommunication companies have tried this route. Verizon and AT&T in the US, for example, ventured into content. Verizon acquired AOL and Yahoo, and AT&T went as far as buying Time Warner in 2018, gaining HBO, Turner, and Warner Bros. with the intent to become a vertically integrated content-and-distribution powerhouse (HBO Max streaming service integrated with AT&T wireless deals). The underlying vertical integration theory was that owning both the infrastructure (wireless/broadband networks) and premium content would create synergies: AT&T could, say, zero-rate HBO Max for its mobile subscribers or use its data on network usage to inform content strategy. However, these moves have had mixed success. AT&T ended up spinning off WarnerMedia in 2022, suggesting that synergy was elusive, and that content and delivery infrastructure might be better separated. Nonetheless, the impulse behind the merger speaks to the recognition that infrastructures and platforms are interdependent.

From the user's perspective, the blending of platform and infrastructure can offer greater efficiency but also raises important concerns about the concentration of power. When a small number of companies control nearly every layer of the digital media ecosystem, the risks of anti-competitive behavior and systemic vulnerability grow. Regulators in Western countries have begun responding to this dynamic. The European Union's Digital Markets Act directly targets dominant firms such as Google, Apple, Amazon, Meta, and Microsoft. The legislation imposes obligations intended to reduce gatekeeping power, including requirements that Apple allow alternative app stores and enable greater interoperability across platforms. These measures represent an effort to loosen the grip of vertically integrated digital ecosystems.

Some observers have gone further, suggesting that large digital platforms should be regulated more like public utilities or common carriers, given their infrastructural importance. Castells noted that much of the public sphere

now operates through privately owned networks, meaning that any failure or bias within those systems can have far-reaching social consequences.⁵¹ A few high-profile outages have made this vulnerability clear. During the 2019 YouTube outage and the 2021 Fastly content delivery network disruption, major swaths of internet services became temporarily unavailable. These incidents served as a reminder that everyday media access often depends on a small number of centralized, privately operated systems.

On the other hand, the integration of platforms with robust infrastructure has enabled remarkable innovations and reliability. Streaming a live global event to millions of mobile users concurrently—something that would have been technically unfathomable not long ago—is now almost routine, owing to these architectures. The COVID-19 pandemic stress-tested the system in 2020, when home and mobile streaming surged. Networks held up generally, with only minor throttling (Netflix and YouTube temporarily reducing default stream quality in Europe to ease network load). The fact that the internet didn't buckle under unprecedented traffic is credited to continual upgrades by infrastructure providers and smart traffic management by platforms which can adjust bitrates adaptively. In a sense, platforms and infrastructure working in concert proved resilient. This resilience, however, often comes from centralized optimization like when Netflix decided unilaterally to lower bitrates globally for a period of time.

Culturally, the merging of platform and infrastructure means that the ways we create, discover, and distribute content are now deeply embedded in technical systems controlled by a few organizations. Whereas twentieth-century culture industries (film studios, record labels, broadcasters) certainly had concentrated power, today's tech platforms arguably wield an even more pervasive influence because they not only produce or license content but also own the channels and the data. A user watching a Netflix-produced show on a Netflix app delivered via Netflix's CDN on a tablet running Netflix's recommended calibration settings is experiencing a vertically integrated chain with very few external inputs. In essence, our deeply rooted user experience is made by possible by, and confined within, a vertically integrated chain controlled by one company. The platform's knowledge about that consumption (what you watched, when, if you liked it) stays with Netflix. In earlier eras, a TV manufacturer or a cable operator might not know exactly which show you were engrossed in, but now the feedback is precise and instant. This feedback loop enables personalization, but it also leaves the traces of

⁵¹ Castells, "Communication, Power and Counter-Power," 239.

our lives in proprietary silos. The consistent monitoring of behavior has been coined the age of surveillance and is central to economic extraction.⁵² Streaming platforms are a prime example, monetizing not just content but the matrix of its consumption.

Finally, the convergence raises the question of global versus local. Western streaming architectures, led by US-based corporations, have become global infrastructures in their own right. Netflix and YouTube reach into hundreds of countries, often dominating local media distribution. This has prompted some nations to push back, either through regulation (like European quotas for local content on Netflix, or Australia and Canada seeking to have YouTube/Facebook pay news outlets for content) or by fostering local platforms. In China, for example, the government kept out many Western platforms, leading to domestic equivalents (Youku for video, Tencent Video, iQIYI, etc., and a highly developed mobile streaming ecosystem behind the “Great Firewall”). Those Chinese platforms have their own integrated infrastructures; Tencent, for one, has massive cloud infrastructure and a constellation of services. So globally, there is a splintering, where Western platform-infrastructure complexes dominate in many regions, but alternative complexes exist in some markets, raising interesting issues of interoperability and digital sovereignty. The phrase “architectures of mobility” consequently also takes on a geopolitical meaning: Whose architecture, whose infrastructure, and whose platforms will structure mobile media access on a global scale?

Conclusion

Mobile streaming media is the product of a profound synthesis of technological infrastructure and platform organization. In this chapter, we traced how streaming services—from Netflix and Spotify to YouTube, TikTok, Twitch, and beyond—depend on and actively shape the underlying networks and systems that deliver content. This tangled web shows that mobile streaming is not just about smartphones as devices, but about entire platform ecosystems and wireless infrastructures working in tandem to enable new habits of media consumption. Throughout this chapter we’ve shown a progression toward greater integration. As networks became faster and more omnipresent, platforms evolved to leverage that connectivity with on-demand,

⁵² Shoshana Zuboff, *The Age of Surveillance Capitalism: The Fight for a Human Future at the New Frontier of Power* (New York: PublicAffairs, 2019), 3.

personalized services. This is mobile personalization at scale. In turn, the success of those services drove further network innovation and investment.

Streaming platforms may feel like tools of convenience and choice, but they operate within a much larger system driven by data extraction, algorithmic control, and economic incentives. These platforms are not just distributing content; they are actively shaping how we experience media. By tracking behavior, prioritizing engagement, and curating what users see, platforms play a central role in influencing attention, tastes, and even public discourse. What appears to be mobile personalization and user empowerment is often engineered for maximum profitability and platform growth.

The shift from traditional broadcasting to on-demand streaming represents more than just a technological upgrade. It reflects a broader transformation in how media systems function—favoring real-time feedback, user profiling, and content optimization over editorial judgment or public service. This has consequences not only for what we watch, but also for how culture circulates, how information is valued, and who gets to be heard.

Understanding streaming through this lens reveals that it's not just about entertainment. It's about infrastructure, power, and control. The platforms that dominate today are shaping the future of media in ways that often go unnoticed. Recognizing this helps us move beyond surface-level appreciation and toward a more critical awareness of how digital media environments are built, maintained, and monetized. In short, mobile streaming's architecture has reshaped media power dynamics: Personalization for the user is built on vast concentrations of power for the platform.

On a technical level, this chapter revealed that the material architecture of mobile streaming is layered and complex. It comprises physical elements (cell towers, servers, etc.), logical protocols (internet protocols, streaming protocols), software systems (CDNs, databases, AI algorithms), and user-facing applications—all coordinated to deliver a smooth experience. Each layer has its guardians: telecom companies, device manufacturers, platform operators, developers, sometimes all within the same corporate entity. The modularity provided by APIs is what allows these layers to interact and the entire edifice to function globally, but it is also what allows certain players to dominate multiple layers by setting the rules of interconnection.

Crucially, mobile streaming's architecture has reconfigured media power dynamics. Users have gained the ability to access a dizzying array of content on their own terms, fragmenting the old mass audiences of the broadcast era into personalized micro-audiences. Yet paradoxically, this individual

empowerment is delivered by highly centralized services—a few recommendation algorithms determining global viewing patterns, a few app stores controlling app distribution, a few network providers carrying the data. The promise of the early internet as a decentralized, democratized media space seems to have been a pipedream. The reality of mobile streaming is more akin to a network of centralized convenience. This does not negate the real benefits—we truly do have more choice and creative opportunity; anyone can start a YouTube or TikTok channel and potentially reach millions—but it contextualizes them to the platform and centralized expectations of the network they are operating within.

Looking ahead, the merging of platforms and infrastructure is likely to deepen with trends like metaverse visions and with technologies like the internet of things and 5G enabling new mobile media forms (immersive and interactive experiences in real physical space). The chapter's insights suggest that unless there are significant changes (regulatory or market-driven), the same few actors will extend their architectures to these new domains, embedding their data-driven, platform-centric logics into the next generation of media systems. However, there is also increasing awareness and pushback: antitrust cases, public pressure on privacy, and open-source or decentralized alternatives. For example, the ActivityPub protocol used by PeerTube for video or Funkwhale for audio aim to be “Fediverse” versions of YouTube/Spotify and seek to reimagine streaming without centralized monopolies. Whether these can offer viable alternatives remains to be seen.

The architectures of mobility that support streaming media are a marvel of modern engineering and enterprise, fusing brick-and-mortar data centers with satellites and subterranean cables with algorithms and attention economies. They represent a new kind of media infrastructure—one that is global, constantly evolving, and intimately entwined with our personal lives. Understanding these architectures is essential. As citizens and consumers, knowing how our media is delivered and shaped allows us to better navigate the contemporary media landscape. This chapter has provided an analysis of those underpinning structures and systems, shedding light on how platform business models, wireless infrastructures, big data, algorithms, and APIs all intersect to produce the mobile streaming experiences that many of us now take for granted. Armed with this understanding, we can better engage in discussions about the future of media, ensuring that the future balances innovation and convenience with fairness, openness, and respect for the human element at its core.